

7.7 Approximate Integration

Suppose we wish to integrate functions for which we do not yet have the techniques, e.g.

$$\int_a^b \frac{e^x}{x} dx, \quad \int_a^b \sin(x^2) dx, \quad \int_a^b \frac{1}{\ln(x)} dx, \quad \int_a^b \sqrt{x^3 + 1} dx, \quad \text{etc.}$$

When we cannot find an antiderivative for a function, we can use approximation techniques to estimate the value of a definite integral.

Review of Approximations

Consider a function $f(x)$ defined on the interval $I = [a, b]$. Divide I into n subintervals of equal width,

$$\Delta x =$$

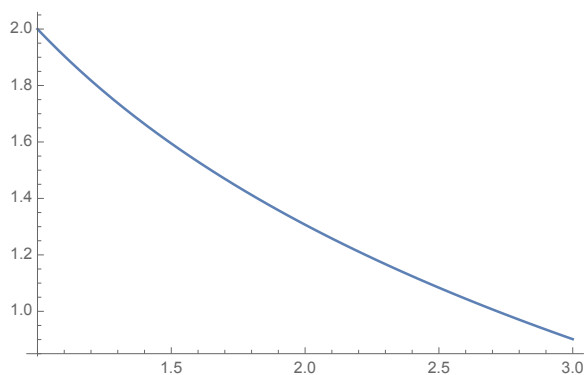
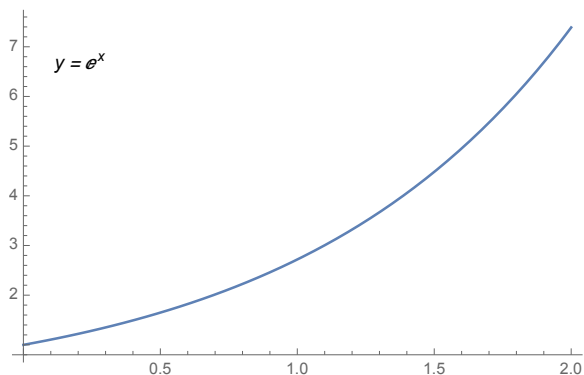
Define $x_i = a + i\Delta x$ as the partition points of the subintervals and we have the three approximations

Left Endpoint Approximation

Right Endpoint Approximation

$$\int_a^b f(x) dx \approx L_n =$$

$$\int_a^b f(x) dx \approx R_n =$$



$f(x)$	L_n	R_n
increasing		
decreasing		

Midpoint Approximation

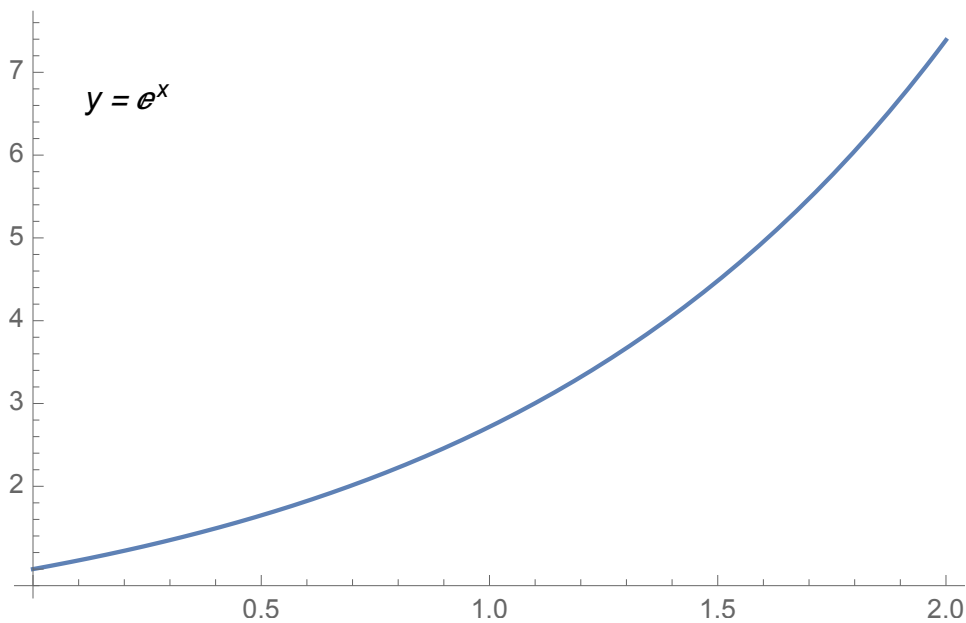
The midpoint approximation for the integral is given as

$$\int_a^b f(x)dx \approx M_n =$$

where $\bar{x}_i =$ _____ for $i = 1, \dots, n$.

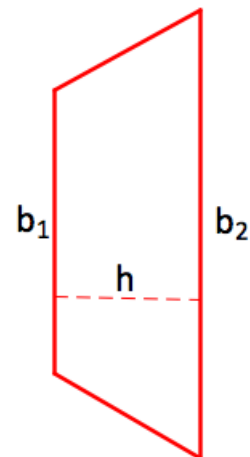
Example 1

Draw L_2 , M_2 , and R_2 on the graph below:

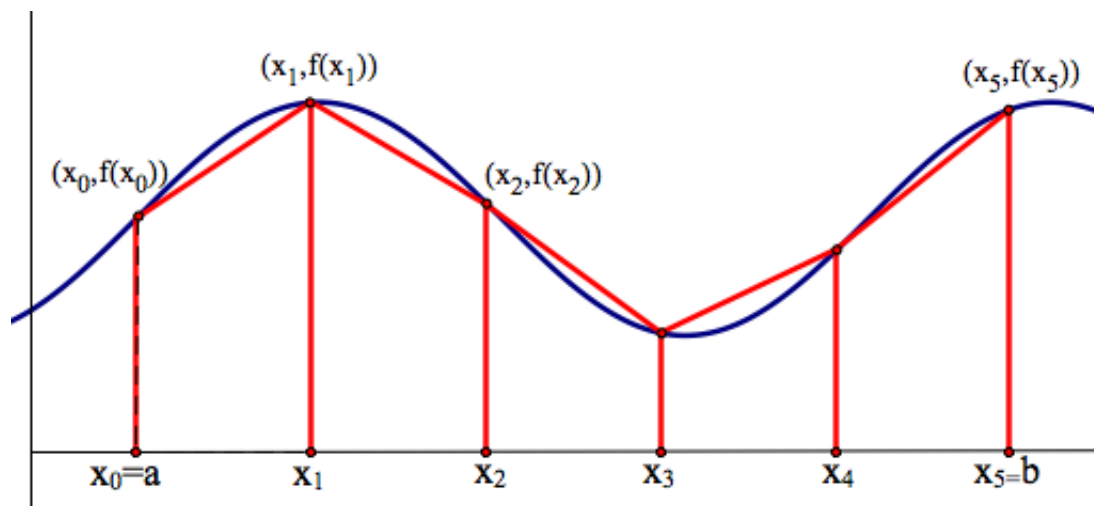


Trapezoidal Rule

How do we find the area of trapezoid?



Given a function $f(x)$ on an interval $[a, b]$, divide the function up on $n = 5$ sub-intervals of equal length and approximate the area with trapezoids.



Derive a formula to find the area under the curve $y = f(x)$ on $[a, b]$ using trapezoids and n sub-intervals.

Trapezoidal Rule

To approximate $\int_a^b f(x) \, dx$, use

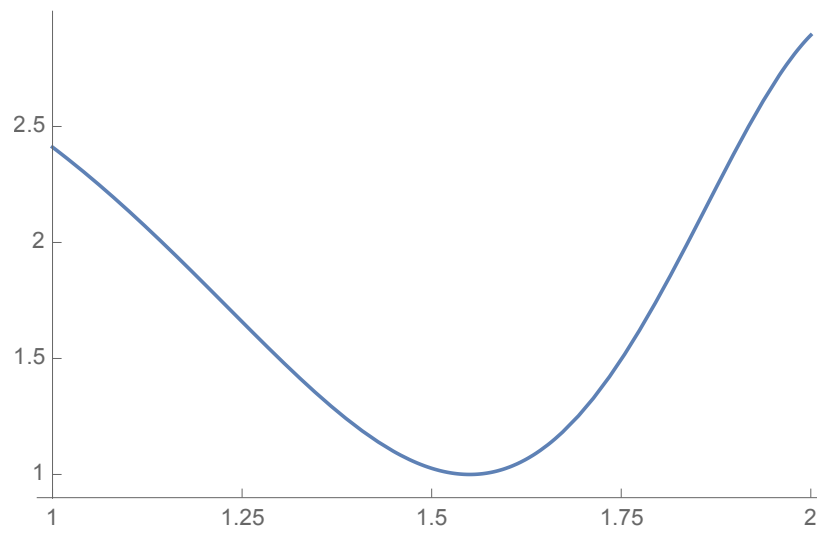
$$\text{where } \Delta x = \frac{b - a}{n}$$

What is the relationship between T_n and R_n ? L_n ?

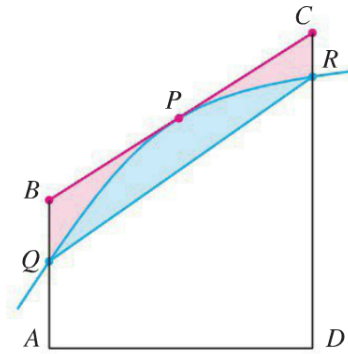
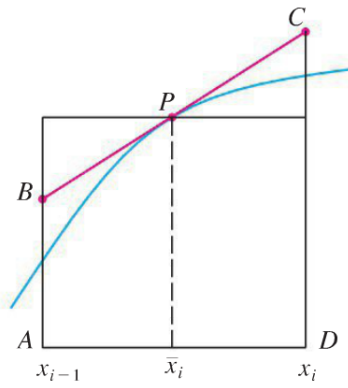
Example 2

Use the Trapezoidal Rule with $n = 4$ to estimate $\int_1^2 (\sin(e^x) + 2) \, dx$.

x	$f(x)$	multiplier	total



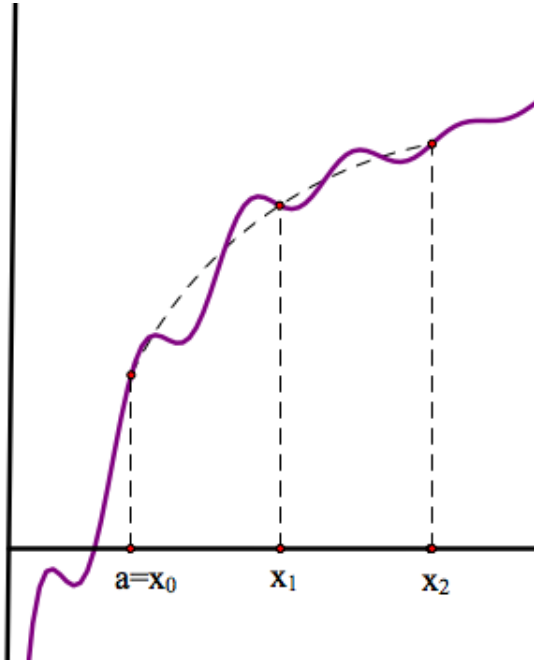
Which is more accurate? M_n or T_n ?



$f(x)$	Graph	T_n	M_n
concave up			
concave down			

Simpson's Rule

Simpson's Rule uses parabolas to approximate the area under the curve.



How many points are necessary to describe a parabola uniquely?

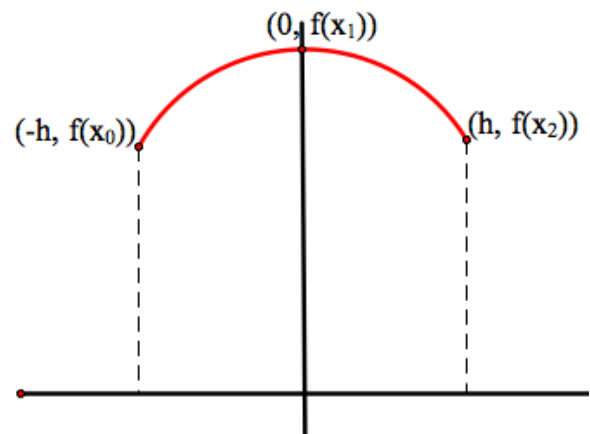
Since Simpson's Rule requires _____, it creates _____ subintervals.

Therefore, the number of subintervals used in Simpson's Rule must be _____.

In order to create an equation for the area under the curve of these parabolae, we must consider facts:

- the equation of a parabola is

- $g(-h) =$ _____
 $g(0) =$ _____
 $g(h) =$ _____



We will now derive a formula for Simpson's Rule:

Step 1: Integrate $\int_{-h}^h (Ax^2 + Bx + C) dx$

Step 2: Use our 3 initial conditions to rewrite the equation.

Step 3: Manipulate equations

Simpson's Rule

To approximate $\int_a^b f(x) dx$, use

where $\Delta x = \frac{b-a}{n}$ and n is even.

Simpson's Rule can also be viewed as a weighted average of the Trapezoidal and Midpoint Rules:

$$S_{2n} = \frac{T_n + M_n + M_n}{3} = \frac{1}{3}T_n + \frac{2}{3}M_n.$$

Since M_n and T_n are expected to have errors with opposite signs, it makes sense to combine them. Since the error bounds for M_n are expected to be half that of T_n it makes sense to weight M_n proportionately more heavily.

What does this mean about the accuracy of Simpson's Rule?

Example 3

Use the Simpson's Rule with $n = 4$ to estimate $\int_0^1 (\sin(x) + \cos(x)) \, dx$.

x	$f(x)$	multiplier	total

Example 4

Using the following table of values, approximate $\int_{-3}^0 f(x) dx$

x	-3	-2.5	-2	-1.5	-1	-0.5	0
$f(x)$	2	1	1	0	1	2	3

- using Trapezoidal Rule

- using Simpson's Rule